

Crossover from Discrete to Continuous Quantum Geometry: Running Spectral Dimensions, Noncommutative Spectral Triples, and Critical Phase Transitions

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

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Abstract

The spectral dimension $d_s(\tau) = -2 \mathrm{d} \ln P / \mathrm{d} \ln \tau$ of a diffusion process is widely used to diagnose dimensional reduction in quantum gravity. We show that it is exactly twice an equipartition count: $d_s(\tau) = 2\tau \langle \lambda \rangle_\tau$, the Gibbs average of the Laplacian at inverse temperature τ . Two consequences are not always appreciated. First, any discretisation with a bounded spectrum gives $d_s \rightarrow 0$ as $\tau \rightarrow 0$, an artefact of the cutoff rather than a feature of quantum geometry. Second, a finite graph gives $d_s \rightarrow 0$ as $\tau \rightarrow \infty$.

For the hypercubic lattice we obtain the exact running dimension $d_s = 2dx [1 - I_1(x)/I_0(x)]$, with $x = 2\tau/a^2$. It rises from zero, overshoots the topological dimension (maximum $1.2178d$) and approaches d from above as $d(1 + a^2/8\tau)$. For the higher-derivative dispersion $E = k^2(1 + \ell^2 k^2)$ in four dimensions, which is the scaling relevant to asymptotic safety, Hořava gravity and quadratic gravity, we give the global interpolation in closed form, $P \propto \tau^{-1} [1 - \sqrt{\pi} s e^{s^2} \operatorname{erfc}(s)]$ with $s = \sqrt{\tau}/2\ell$, which runs from $d_s = 2$ to $d_s = 4$. Dimensional splitting $d_s \neq d_H$ is traced to an anomalous walk dimension, $d_s = 2d_H/d_w$. On the Sierpinski gasket this gives $d_s = 2 \ln 3 / \ln 5$, with log-periodic oscillations of period $\ln 5$.

On the noncommutative side, we prove that the lattice Hodge–Dirac operator $d + d^*$ on the torus T^d converges to the continuum Hodge–Dirac operator in norm-resolvent sense with error $O(a)$, whereas the naive symmetric-difference Dirac operator does not converge, because of fermion doubling. For the incidence Dirac operator of a graph, the Connes distance is bounded between $a d_{\text{hop}} / \sqrt{d_{\text{deg}}}$ and $a d_{\text{hop}}$. On $\mathbb{Z}^d$ the hop metric tends to the ℓ^1 metric, not the Euclidean one; nevertheless, numerical lower bounds on the Connes distance converge to the Euclidean distance.

All statements are checked numerically by exact diagonalisation.

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1 Introduction

Several approaches to quantum gravity find that the effective dimension of spacetime depends on the scale at which it is probed, and that it decreases towards two at short distances. Examples include causal dynamical triangulations (CDT) [1], asymptotic safety [2], Hořava–Lifshitz gravity [3], loop quantum gravity and spin foams [4], quantum spacetimes with fractal properties [5] and multifractional theories [6]; see [7] for a review. The usual diagnostic is the spectral dimension

$$d_s(\tau) = -2 \frac{d \ln P(\tau)}{d \ln \tau}, \quad P(\tau) = \frac{1}{V} \text{Tr } e^{-\tau \Delta}, \quad (1)$$

defined by the return probability of a diffusion process. Different definitions of the diffusion can give different answers [8], and the interpretation of d_s requires care.

This paper studies the crossover from discrete to continuous geometry in three complementary languages: heat kernels, noncommutative spectral triples [9, 10], and fractal graphs. Our main results are the following.

1. *Spectral dimension as equipartition* (Theorem 1). $d_s(\tau) = 2\tau \langle \lambda \rangle_\tau$ exactly. As consequences, the ultraviolet value $d_s \rightarrow 0$ found for any discretisation is a cutoff artefact, a finite graph has $d_s \rightarrow 0$ also in the infrared, and plateaus of d_s correspond to power laws in the spectral density.
2. *Exact running dimensions* (Propositions 2.1 and 2.2). For the hypercubic lattice the running is non-monotonic and overshoots d . For the dispersion $E = k^2(1 + \ell^2 k^2)$ we give a closed-form global interpolation from 2 to 4 in $d = 4$, and a general- d formula in terms of parabolic-cylinder functions.
3. *Dimensional splitting* (Section 3). $d_s \neq d_H$ arises from an anomalous walk dimension, $d_s = 2d_H/d_w$ [11, 12]. In quantum gravity this corresponds to an anomalous scaling of the kinetic operator at fixed volume scaling. Self-similar graphs add log-periodic oscillations [13].
4. *Spectral triples* (Section 4). We prove $O(a)$ norm-resolvent convergence of the lattice Hodge–Dirac operator on T^d , show that the naive lattice Dirac operator fails to converge because of fermion doubling [14], and bound the Connes distance on graphs.

We also correct three statements that are sometimes made about this crossover. A vanishing ultraviolet spectral dimension is generic for any discretisation and carries no information about quantum gravity. The hop metric of a hypercubic lattice converges to the ℓ^1 metric, not to the Riemannian distance. Lattice corrections to heat-kernel coefficients are non-universal ultraviolet terms, not conformal anomalies.

This is the fourth paper of the author’s programme of scale-dependent analysis (“pan-scale analytics”), which also covers complex dimensions in holographic entanglement [19], multifractal entanglement spectra [20], and algorithmic complexity under scaling [21]. Connections are discussed in Section 5.

2 The spectral dimension as an equipartition count

2.1 General identity

Theorem 1 (Equipartition form of the spectral dimension). *Let $\Delta \geq 0$ have spectral measure ρ with $P(\tau) = \int e^{-\tau\lambda} d\rho(\lambda) < \infty$ for $\tau > 0$. Then*

$$d_s(\tau) = 2\tau \langle \lambda \rangle_\tau, \quad \langle \lambda \rangle_\tau = \frac{\int \lambda e^{-\tau\lambda} d\rho(\lambda)}{\int e^{-\tau\lambda} d\rho(\lambda)}. \quad (2)$$

In words, $d_s/2$ is the mean energy, in units of the temperature $1/\tau$, of a canonical ensemble whose levels are the eigenvalues of Δ .

Proof. Differentiate under the integral: $-\tau d \ln P / d\tau = \tau \int \lambda e^{-\tau\lambda} d\rho / \int e^{-\tau\lambda} d\rho$. $\square$

For a continuum Laplacian, $\rho(\lambda) \propto \lambda^{d/2-1}$, and each of the d quadratic degrees of freedom contributes $1/(2\tau)$ to $\langle \lambda \rangle_\tau$, which gives $d_s = d$. This is equipartition. The same thermodynamic reading of heat-kernel and Rényi quantities underlies the companion paper [20].

Corollary 2.1. (i) *If $\text{spec } \Delta \subset [0, \lambda_{\max}]$, then $0 \leq d_s(\tau) \leq 2\tau\lambda_{\max}$, and so $d_s \rightarrow 0$ as $\tau \rightarrow 0$.*

(ii) *On a finite connected graph, $d_s(\tau) \leq 2\tau(N-1)\lambda_{\max}e^{-\tau\lambda_1}$, where λ_1 is the smallest non-zero eigenvalue. Hence $d_s \rightarrow 0$ as $\tau \rightarrow \infty$.*

(iii) *If $\rho([0, \lambda]) \asymp \lambda^{\delta/2}$ for $\lambda_- \ll \lambda \ll \lambda_+$, then $d_s(\tau) \approx \delta$ for $1/\lambda_+ \ll \tau \ll 1/\lambda_-$.*

Proof. (i) Use $\langle \lambda \rangle_\tau \leq \lambda_{\max}$. (ii) The zero mode has weight $1/N$ in P and contributes nothing to the numerator, which is at most $(N-1)\lambda_{\max}e^{-\tau\lambda_1}/N$. (iii) In the scaling window the Gibbs average of a power-law density gives $\langle \lambda \rangle_\tau = \delta/(2\tau)$, with corrections from the edges of the window suppressed exponentially or by powers of $\tau\lambda_{\pm}$. $\square$

Part (i) has a direct implication for quantum gravity. A vanishing spectral dimension at the shortest scales is produced by *any* regularisation with a bounded spectrum, so it cannot by itself be evidence of new physics. Physically meaningful ultraviolet values, such as $d_s \approx 2$, must appear as plateaus at intermediate τ , well above the cutoff scale.

2.2 The hypercubic lattice

Proposition 2.1 (Exact lattice running). *For the graph Laplacian of $a\mathbb{Z}^d$, with dispersion $E(k) = \sum_i \frac{2}{a^2}(1 - \cos k_i a)$, the return probability per site is $P(\tau) = [e^{-x}I_0(x)]^d$ with $x = 2\tau/a^2$, and*

$$d_s(\tau) = 2dx \left(1 - \frac{I_1(x)}{I_0(x)}\right), \quad x = \frac{2\tau}{a^2}. \quad (3)$$

As $\tau \rightarrow 0$, $d_s = 2dx + O(x^2)$. As $\tau \rightarrow \infty$,

$$P(\tau) = (4\pi\tau)^{-d/2}a^d \left(1 + \frac{d a^2}{16\tau} + O(a^4/\tau^2)\right), \quad d_s(\tau) = d \left(1 + \frac{a^2}{8\tau} + O(a^4/\tau^2)\right). \quad (4)$$

Consequently d_s approaches d from above. Its maximum is $d_s^{\max} = 1.2178d$, attained at $\tau \simeq 0.84a^2$.

Proof. The one-dimensional factor is $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-x(1-\cos\theta)} d\theta = e^{-x}I_0(x)$, which gives the product form. Then $\frac{d}{dx} \ln[e^{-x}I_0(x)] = I_1/I_0 - 1$, and (3) follows. The asymptotic forms follow from $I_\nu(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{4\nu^2-1}{8x} + \dots\right)$ [17], which gives $1 - I_1/I_0 = \frac{1}{2x} + \frac{1}{8x^2} + O(x^{-3})$. The maximum was located numerically from (3). $\square$

The a^2/τ term is a non-universal ultraviolet correction to the heat-kernel coefficients. It is not a conformal anomaly: the anomaly is controlled by the coefficient $a_{d/2}$ of the Seeley–DeWitt expansion, which is a curvature invariant and vanishes on a flat torus.

2.3 Higher-derivative dispersion: a closed-form global interpolation

Effective dispersions $E \propto k^2(1 + \ell^2 k^2)$ arise from the transverse-traceless propagator of quadratic gravity [15]. Their ultraviolet scaling $E \propto k^4$ reproduces $d_s = 2$ in four dimensions, as found in asymptotic safety from the anomalous dimension $\eta_N = -2$ [2] and in Hořava gravity with $z = 3$ [3].

Proposition 2.2 (Global interpolation). *For $E(k) = k^2(1 + \ell^2 k^2)$ in $\mathbb{R}^d$ with flat measure,*

$$P(\tau) \propto \int_0^\infty u^{\frac{d}{2}-1} e^{-\tau u - \tau \ell^2 u^2} du = \Gamma(\frac{d}{2}) (2\tau \ell^2)^{-d/4} e^{\tau/(8\ell^2)} D_{-d/2}\left(\frac{\sqrt{\tau}}{\sqrt{2}\ell}\right), \quad (5)$$

with D_ν the parabolic-cylinder function [18]. For $d = 4$ this reduces to

$$P(\tau) \propto \frac{1}{\tau} \left[1 - \sqrt{\pi} s e^{s^2} \operatorname{erfc}(s) \right], \quad s = \frac{\sqrt{\tau}}{2\ell}, \quad (6)$$

so $d_s \rightarrow 2$ for $\tau \ll \ell^2$ and $d_s \rightarrow 4$ for $\tau \gg \ell^2$. More generally, $d_s = 2d/\alpha$ in a regime where $E \propto k^\alpha$.

Proof. Set $u = k^2$ and use [18, 3.462.1]. For $d = 4$, evaluate $\int_0^\infty u e^{-\beta u^2 - \gamma u} du$ directly, with $\beta = \tau \ell^2$ and $\gamma = \tau$. The limits follow from $\sqrt{\pi} s e^{s^2} \operatorname{erfc}(s) = 1 - \frac{1}{2s^2} + O(s^{-4})$ as $s \rightarrow \infty$, and from $\rightarrow 0$ as $s \rightarrow 0$. For a pure power law $E \propto k^\alpha$, Theorem 1 gives $\langle E \rangle_\tau = d/(\alpha\tau)$. $\square$

Combining (6) with a momentum cutoff $|k| \leq \pi/a$ gives three regimes: $d_s \rightarrow 0$ for $\tau \ll a^4/\ell^2$, $d_s \approx 2$ in between, and $d_s \rightarrow 4$ for $\tau \gg \ell^2$. Numerically, with $\ell = 10a$, the cutoff result agrees with (6) to 10^{-4} for $\tau \geq 10^{-2}a^2$. It equals 0.001 at $\tau = 10^{-7}a^2$, averages 2.03 on $10^{-2} \leq \tau/a^2 \leq 1$, and reaches 3.999 at $\tau = 10^6a^2$. We also checked the parabolic-cylinder identity (5) numerically to 10^{-11} (Fig. 1, bottom right).

3 Fractal graphs and dimensional splitting

3.1 Walk dimension

Let d_H be the Hausdorff (volume-growth) dimension and d_w the walk dimension, defined by $\langle r^2 \rangle \sim \tau^{2/d_w}$. Then [11, 12]

$$d_s = \frac{2d_H}{d_w}. \quad (7)$$

Dimensional splitting, $d_s < d_H$, is thus equivalent to subdiffusion, $d_w > 2$. For a dispersion $E \propto k^\alpha$ on a space with ordinary volume growth $d_H = d$, the diffusion is anomalous with $d_w = \alpha$, and (7) gives $d_s = 2d/\alpha$, consistent with Proposition 2.2. This makes precise how quantum-gravity proposals can combine $d_H \approx 4$ with $d_s \approx 2$. The splitting reflects an anomalous scaling of the kinetic operator ($\alpha = 4$) at unchanged volume scaling. It does not require the geometry to be fractal in the sense of volume growth.

3.2 Sierpinski gasket

For the Sierpinski gasket, $d_H = \ln 3 / \ln 2$ and $d_w = \ln 5 / \ln 2$, so $d_s = 2 \ln 3 / \ln 5 \simeq 1.3652$ [12]. Spectral decimation rescales eigenvalues by 5 under the geometric scaling by 2. As a result $P(\tau) = \tau^{-d_s/2} G(\ln \tau)$ with G periodic of period $\ln 5$, and $d_s(\tau)$ oscillates log-periodically. This is the heat-kernel counterpart of the complex dimensions studied in [13, 19].

Numerical test. We diagonalised the graph Laplacian of the level-7 gasket ($N = 3282$ vertices). The finite-size scale is $1/\lambda_1 \simeq 4.3 \times 10^3$. Averaged over exactly three periods, d_s is 1.387, 1.357 and 1.347 for $\tau \in [1, 125]$, $[2, 250]$ and $[3, 375]$, bracketing $2\ln 3/\ln 5$. The drift reflects the ultraviolet transient on the left and the onset of finite-size effects on the right. After linear detrending, the dominant oscillation frequency is 1.02 cycles per unit of $\log_5 \tau$, against the predicted value 1 (Fig. 2, left).

3.3 Critical phase transitions and effective functionals

In CDT, macroscopic four-dimensional geometry emerges in one phase of the path integral, and a second-order transition between phases has been reported [16], which allows a continuum limit to be considered. A natural effective description of the near-critical regime, consistent with the scale dependence of d_s , is a higher-curvature functional of the form

$$\Gamma_k[g] = \int d^4x \sqrt{g} \left[\frac{1}{16\pi G_k} (R - 2\Lambda_k) + \alpha_k R^2 + \beta_k C_{\mu\nu\rho\sigma} C^{\mu\nu\rho\sigma} \right], \quad (8)$$

with scale-dependent couplings. On flat space its transverse-traceless kinetic operator has the form $k^2(1 + \ell^2 k^2)$ with $\ell^2 \propto \beta G$ [15], which is the dispersion of Proposition 2.2. We stress that (8) is a truncation motivated by these properties. We do not derive it from the CDT transition, whose universality class remains open.

4 Noncommutative spectral triples: from lattice to continuum

4.1 Norm-resolvent convergence of the lattice Hodge–Dirac operator

Let $T^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ and $a = 2\pi/N$. The cubical lattice of spacing a carries discrete differential forms, the coboundary d_a and its adjoint, and the Hodge–Dirac operator $D_a = d_a + d_a^*$ on $\mathcal{H}_a = \bigoplus_p \Lambda_a^p$. We use the spectral triple $(C(\text{lattice}), \mathcal{H}_a, D_a)$. The continuum counterpart is $D = d + d^*$ on $L^2\Lambda^\bullet(T^d)$. Let J_a identify the lattice Fourier mode k in the Brillouin zone $B_a = (-\pi/a, \pi/a]^d$ with the corresponding continuum mode and form component.

Theorem 2 (Norm-resolvent convergence). *For every $z \in \mathbb{C} \setminus \mathbb{R}$ there is $C(z)$ such that*

$$\|J_a(D_a - z)^{-1}J_a^* - (D - z)^{-1}\| \leq C(z) a. \quad (9)$$

Proof. By the Künneth structure of the cubical complex of T^d , D_a^2 acts on each form component as the scalar lattice Laplacian. Hence D_a has eigenvalues $\pm s_a(k)$, with $s_a(k)^2 = \sum_i \frac{4}{a^2} \sin^2 \frac{k_i a}{2}$ for $k \in B_a$, while D has eigenvalues $\pm |k|$ with the same multiplicities. For $|k_i a| \leq \pi$ we have $\frac{2}{\pi} |k_i| \leq \frac{2}{a} |\sin \frac{k_i a}{2}| \leq |k_i|$ and $0 \leq |k_i| - \frac{2}{a} |\sin \frac{k_i a}{2}| \leq a^2 |k_i|^3/24$. Therefore $\frac{2}{\pi} |k| \leq s_a(k) \leq |k|$ and $|k| - s_a(k) = \frac{|k|^2 - s_a^2}{|k| + s_a} \leq a^2 |k|^3/12$. Take $z = iy$ for simplicity. For a mode in B_a ,

$$\left| \frac{1}{\pm s_a - z} - \frac{1}{\pm |k| - z} \right| = \frac{||k| - s_a|}{|s_a \mp z| | |k| \mp z|} \leq \frac{a^2 |k|^3/12}{s_a |k|} \leq \frac{\pi a^2 |k|}{24} \leq \frac{\pi^2 \sqrt{d}}{24} a, \quad (10)$$

since $|k| \leq \pi\sqrt{d}/a$ in B_a . For modes outside B_a , which lie in the kernel of J_a^* , the continuum resolvent is bounded by $1/|k| \leq a/\pi$. Both operators are diagonal in the Fourier basis, so the norm is the supremum of these bounds. General z follows from the resolvent identity. $\square$

Counterexample: fermion doubling. The naive Dirac operator $\sum_\mu \gamma^\mu \nabla_\mu^{\text{sym}}$ has eigenvalues $\pm (\sum_\mu \sin^2(k_\mu a)/a^2)^{1/2}$, which vanish at every corner $k_\mu \in \{0, \pi/a\}$ of the Brillouin zone. At $k = (\pi/a, 0, \dots)$ the lattice resolvent equals $-1/z$, whereas the continuum resolvent at $|k| = \pi/a$ is $O(a)$. The difference therefore stays of order one, and there is no norm-resolvent convergence. This is the Nielsen–Ninomiya obstruction [14]. Numerically, on the circle at $z = i$, the Hodge–Dirac error decreases as 1.4×10^{-1} , 3.2×10^{-2} , 7.9×10^{-3} , 2.0×10^{-3} , 4.9×10^{-4} for $N = 16, \dots, 4096$, i.e. as $O(a)$. The naive error stays at 1.00 (Fig. 2, middle).

4.2 Connes distance on graphs

The Connes distance is $d_C(p, q) = \sup\{|f(p) - f(q)| : \|[D, f]\| \leq 1\}$ [9, 10]. For a graph with edges oriented and weighted by $1/a$, take the incidence Dirac operator $D = \begin{pmatrix} 0 & \partial^* \\ \partial & 0 \end{pmatrix}$ on $\ell^2(V) \oplus \ell^2(E)$, with $(\partial g)(e) = (g(e^+) - g(e^-))/a$. Functions act on edges by $f(e) = f(e^-)$.

Proposition 4.1 (Connes distance bounds). *Let $\deg^+$ be the maximal in-degree. Then $\|[D, f]\|^2 = \max_v \sum_{e:e^+=v} |\nabla_e f|^2$ with $\nabla_e f = (f(e^+) - f(e^-))/a$, and*

$$\frac{a d_{\text{hop}}(p, q)}{\sqrt{\deg^+}} \leq d_C(p, q) \leq a d_{\text{hop}}(p, q). \quad (11)$$

In one dimension, with consistent orientation, $d_C(p, q) = |p - q|$ exactly.

Proof. $[\partial, f]g(e) = \nabla_e f g(e^+)$ is a weighted map from vertices to edges, and its norm is $\max_v (\sum_{e^+=v} |\nabla_e f|^2)^{1/2}$. The same holds for $[\partial^*, f]$, which is minus its adjoint. Hence $\|[D, f]\| \leq 1$ implies $|\nabla_e f| \leq 1$ on every edge, so $d_C \leq a d_{\text{hop}}$. Conversely, $f = a d_{\text{hop}}(p, \cdot)/\sqrt{\deg^+}$ is admissible. In one dimension $\deg^+ = 1$. $\square$

On $a\mathbb{Z}^d$, $a d_{\text{hop}}$ converges to the ℓ^1 distance, not to the Euclidean distance; the ratio is $\sqrt{2}$ along a diagonal of $\mathbb{Z}^2$. The hop metric therefore does not deform into the Riemannian geodesic distance. The Connes distance is subtler. On $\mathbb{Z}^2$, with edges oriented along $+x$ and $+y$, the upwind recursion $f(v) = \frac{A+B}{2} + \frac{1}{2}\sqrt{2 - (A-B)^2}$, where A and B are the values at the two predecessors, defines an admissible f . It is a monotone discretisation of the eikonal equation $|\nabla f| = 1$ and gives a lower bound on d_C . Relative to the Euclidean distance, this bound equals 1.0186, 1.0063 and 1.0020 at 45° for grids of 50, 200 and 800 steps, and exactly 1 along the axes (Fig. 2, right).

Conjecture 1. *For the incidence Dirac operator on $a\mathbb{Z}^d$, $d_C(p, q) \rightarrow |p - q|_2$ as $a \rightarrow 0$.*

The lower bound converges to $|p - q|_2$. An upper bound sharper than ℓ^1 is missing.

5 Discussion and limitations

1. The identity $d_s = 2\tau\langle\lambda\rangle_\tau$ shows that the spectral dimension is a property of the spectral density alone. It is insensitive to eigenvectors, and hence to geometric information beyond the spectrum. Two geometries with the same spectrum have the same running d_s .
2. The ultraviolet zero and the infrared drop in Corollary 2.1 are generic artefacts. Reported ultraviolet values of d_s should be read from plateaus well above the cutoff scale and below the finite-size scale.
3. Theorem 2 concerns flat tori. Extending norm-resolvent convergence to curved manifolds and random triangulations, as in CDT, and controlling fermion doubling there, are open problems.
4. The conjectured Euclidean limit of the Connes distance on $\mathbb{Z}^d$ is supported by lower bounds only.
5. The functional (8) is a truncation, not a derivation from the CDT transition.
6. *Connections within the series.* Log-periodic oscillations of d_s on self-similar graphs are the heat-kernel counterpart of the complex dimensions of [19]. The equipartition form of d_s is an instance of the thermodynamic formalism used for entanglement spectra in [20]. Whether a computable graph can have an uncomputable spectral dimension, in analogy with the uncomputable fractal dimensions of [21], is left open.

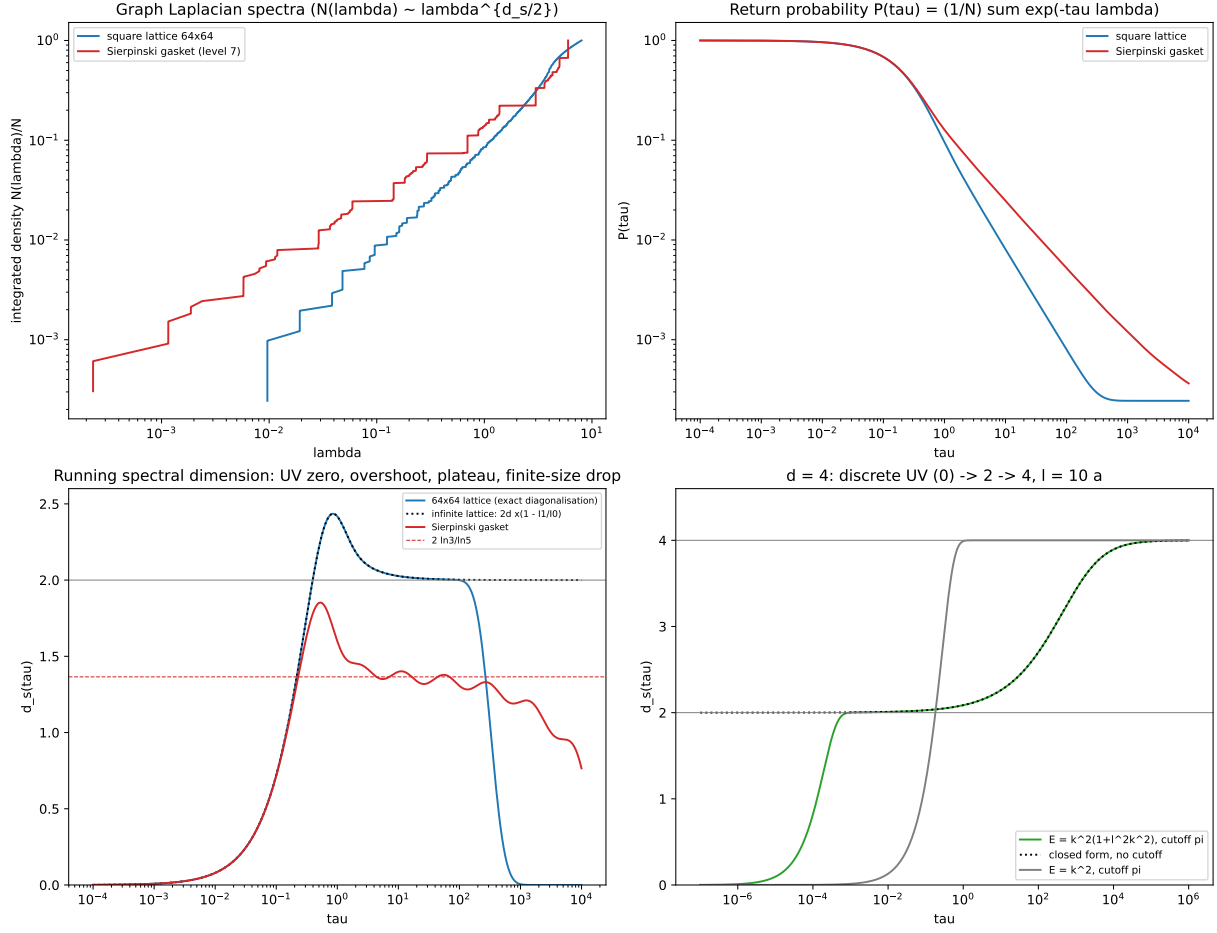


Figure 1: Top left: integrated spectral densities of the 64×64 periodic lattice and the level-7 Sierpinski gasket. Top right: return probabilities. Bottom left: running spectral dimension. The lattice (exact diagonalisation) coincides with the infinite-lattice formula (3) until finite-size effects set in near $\tau \sim 10^2$; note the overshoot above $d = 2$. The gasket oscillates around $2 \ln 3 / \ln 5$. Bottom right: $d = 4$ with $E = k^2(1 + \ell^2 k^2)$, $\ell = 10a$, and cutoff $|k| \leq \pi/a$ (green), the closed form (6) (dotted), and $E = k^2$ with the same cutoff (grey).

Data and code availability

All numbers and figures are produced by the Python script `quantum_geometry_crossover.py`, available at <https://github.com/Ruqing1963/quantum-geometry-crossover> and archived on Zenodo under DOI [10.5281/zenodo.23195881](https://doi.org/10.5281/zenodo.23195881).

A Numerical methods

- *Graphs.* The periodic 64×64 square lattice and the level-7 Sierpinski gasket (built recursively on integer coordinates; $N = (3^8 + 3)/2 = 3282$) are diagonalised exactly with LAPACK (`numpy.linalg.eigvalsh`). The lattice eigenvalues agree with the analytic spectrum to 10^{-13} .
- *Heat kernel.* $P(\tau) = \frac{1}{N} \sum_k e^{-\tau \lambda_k}$ on 321 logarithmically spaced points in $[10^{-4}, 10^4]$. d_s is computed both by centred differences in $\ln \tau$ and from (2); the two agree to 1.5×10^{-3} .
- *Modified dispersion.* Radial integrals with cutoff π/a are evaluated by adaptive quadrature (relative tolerance 10^{-11} , no absolute tolerance), split at $k = \tau^{-1/2}$, on $\tau \in [10^{-7}, 10^6]$.

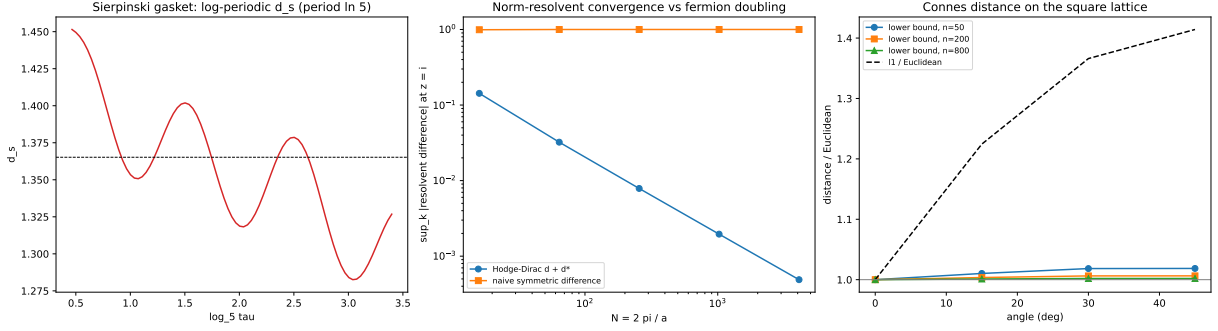


Figure 2: Left: log-periodic oscillation of d_s on the Sierpinski gasket (period $\ln 5$). Middle: norm-resolvent error at $z = i$ on the circle for the lattice Hodge-Dirac operator (convergent, $O(a)$) and the naive symmetric-difference operator (non-convergent, fermion doubling). Right: lower bound on the Connes distance on the square lattice relative to the Euclidean distance, compared with the ℓ^1 distance.

Tails with $\tau E > 700$ are omitted, since they underflow.

- *Dirac operators.* Resolvent differences are evaluated mode by mode on the circle, including the bound $1/|k|$ for continuum modes outside the Brillouin zone.
- *Connes distance.* The upwind recursion is checked to satisfy the admissibility constraint at every vertex; values are reported at the lattice point on each ray at the edge of the grid.

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